



Australia's
Global
University

Faculty of Engineering

School of Electrical Engineering and Telecommunications

Introduction to ELEC1111

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School of Electrical Engineering and Telecommunications, UNSW

About Me

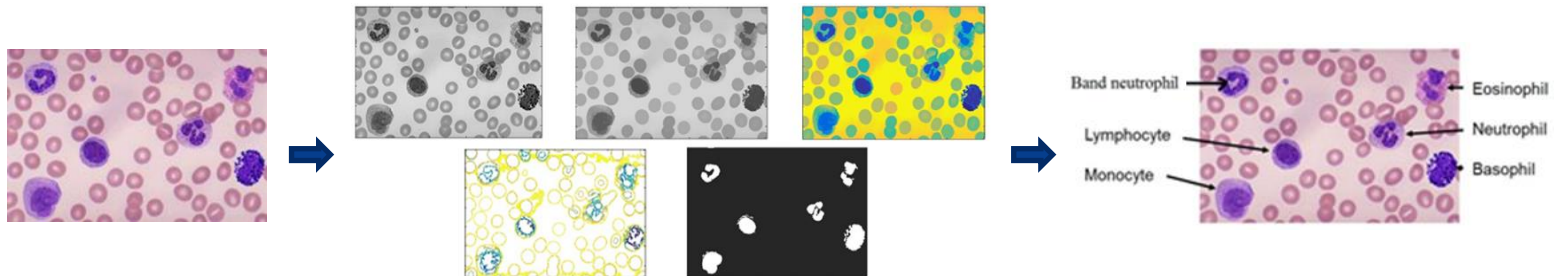
Background

- Senior Lecturer (Education focussed), UNSW, Australia (2018 –).
- Associate Lecturer, QUT, Australia (2015 – 2018).
- PhD in Electrical Engineering, QUT, Australia (2015).
- R&D engineer, UC3M, Spain (2008 – 2011).
- Bachelor (2006) and Master (2008) of Telecommunications Engineering, and Master in Multimedia and Communications (2010), UC3M, Spain.



Main Research Interests

- Signal and Image Processing.
- Pattern Recognition, Machine Learning and AI.
- Engineering Education.



About You

The screenshot shows a web interface for the course 'ELEC1111-Elec & Telecomm Eng - 2024 T3'. On the left is a sidebar with a list of links: 'Quick Links', 'General Course Resources' (highlighted with a red box), 'Assessments Hub', 'Lecture Recordings', and 'Basics and Background'. The main content area has tabs for 'Course', 'Participants', 'Grades', and 'Competencies'. Under the 'Course' tab, there is a 'Getting Started' section with a paragraph of introductory text. Below this are two links: 'Course Outline' and 'Course Information', both preceded by a globe icon. Further down is a 'Pre-course Survey' link, preceded by a speaker icon, which is also highlighted with a red box. Below the survey link is a short description: 'A quick introductory survey for this Term.'

ELEC1111-Elec & Telecomm Eng - 2024 T3

- > Quick Links
- > **General Course Resources**
- > Assessments Hub
- > Lecture Recordings
- > Basics and Background

Course Participants Grades Competencies

Getting Started

Here you will find the basic information of the course, its aim and what is expected of you. Please review this section carefully.

- [Course Outline](#)
- [Course Information](#)

This course information is revised in detail during the first hour of the first lecture.

- [Pre-course Survey](#)

A quick introductory survey for this Term.

Please complete the [survey](#) by the end of this week.

About the Course

Features

- Introductory course in Electrical Engineering.
- Provides an introduction to simple electrical circuits, as well as technical skills to analyse them.
- Hands-on experience in building and testing circuits.
- Co-requisite for Digital Circuit Design (ELEC2141)
Co-requisite for Circuits and Signals (ELEC2134)

Assumed knowledge

- Physics (high-school level).
- Calculus (high-school level).
 - Linear equations and complex numbers*.

* Review material available in Moodle → Basics and Background.

Course Staff

- Teaching team consists of 35 people (plus markers).

Lecturer (content-related queries)

- Dr. Inmaculada (Inma) Tomeo-Reyes
Room 414, EE&T building (G17)
i.tomeo-reyes@unsw.edu.au

Consultation Times

- Wednesdays and Thursdays after the lecture.
- By appointment via email.

Course coordinator and tutor (course-related queries)

- Dr. Wendy Lee
wendy.lee@unsw.edu.au

Laboratory contact person (lab-related queries)

- Ms. Eying Sim Wong
ELEC1111_lab_queries@unsw.edu.au

Ask Away!



Scout

Student Admin Chatbot



What is the purpose?

- Provide course details (from course outline and FAQs received previously in this course)
- Easily accessible university information
- To streamline student support at UNSW



What can this chatbot do?

- Provides quick access to important dates (e.g. "When is census date")
- Course related information (e.g. "what is the weighting of midterm exam?")
- Information on student life (e.g. "Is there a chess club at UNSW?")



What now?

- Talk to the chatbot from your Moodle course site
- We will keep improving the chatbot's ability based on your feedback
- Contact a human support team through the chatbot
- Always verify the info provided by the chatbot
- Do not provide any personal/identifiable info



Scan to
answer a quick
survey
(anonymous)

Ask Away!

Ed Discussion Board

← → ↺ 🏠

🔒 <https://edstem.org/au/courses/19021/discussion/2200949>

ed 

ELEC1111 – Ed Discussion

New Thread

COURSES

ELEC1111

ELEC2134

Drafts

Scheduled

CATEGORIES

General

Lectures

Labs

Quizzes

Workshops

Social

Search

Filter

Welcome to the ELEC1111 Discussion Board

General Wendy Lee LECTURER / COURSE CONVENOR 2d 6

This Week

Electrical Engineering OH&S Topic 3

Labs - 1 Anonymous 19h 3

Purchasing lab kit

General Anonymous 2d 10 2

Welcome to the ELEC1111 Discussion Board #3



Wendy Lee

LECTURER / COURSE CONVENOR

2 days ago in General

UNPIN

STAR

WATCH

85 VIEWS

♡

6

Dear ELEC1111 students,

Welcome to the **ELEC1111 Ed Discussion Board** - This will be the **main forum** to ask questions pertaining course.

Search before you post, in case your question has been asked before.

Use the category that better fits your question (General, Lectures, Labs or Quizzes). There is also a Social section for student interaction. Please be respectful to one another.

If you need additional assistance, attend the drop-in centre sessions or use consultation time with lecturer (details as follows).

Drop-in centre sessions:

- Mondays 3-4pm, [Quadrangle G055](#) (Weeks 2-5, 7-10) - PASS (Peer Assisted Study Session)
- Mondays 4-6pm, [Quadrangle G055](#) (Weeks 2-5, 7-10)
- Tuesdays 6-7pm, [Online](#) (Weeks 2-5, 7-10) - PASS (Peer Assisted Study Session)
- Wednesdays 9-11am, [Quadrangle G055](#) (Weeks 2-5, 7-10)
- Fridays 1-2pm, [Quadrangle G055](#) (Weeks 2-5, 7-10) - PASS (Peer Assisted Study Session)

Consultation times with lecturer:

- Wednesdays & Thursdays after the lecture (for as long as necessary)
- By appointment via email (l.tomeo-reyes@unsw.edu.au) for all other times
 - Location: Electrical Engineering Building (G17), Level 4, Room 414
- For non-content related material, academic advice and personal questions, please reach out to your Course Convenor via email (wendy.lee@unsw.edu.au)
 - Location: Electrical Engineering Building (G17), Level 3, Room 312

Comment Edit Delete ...

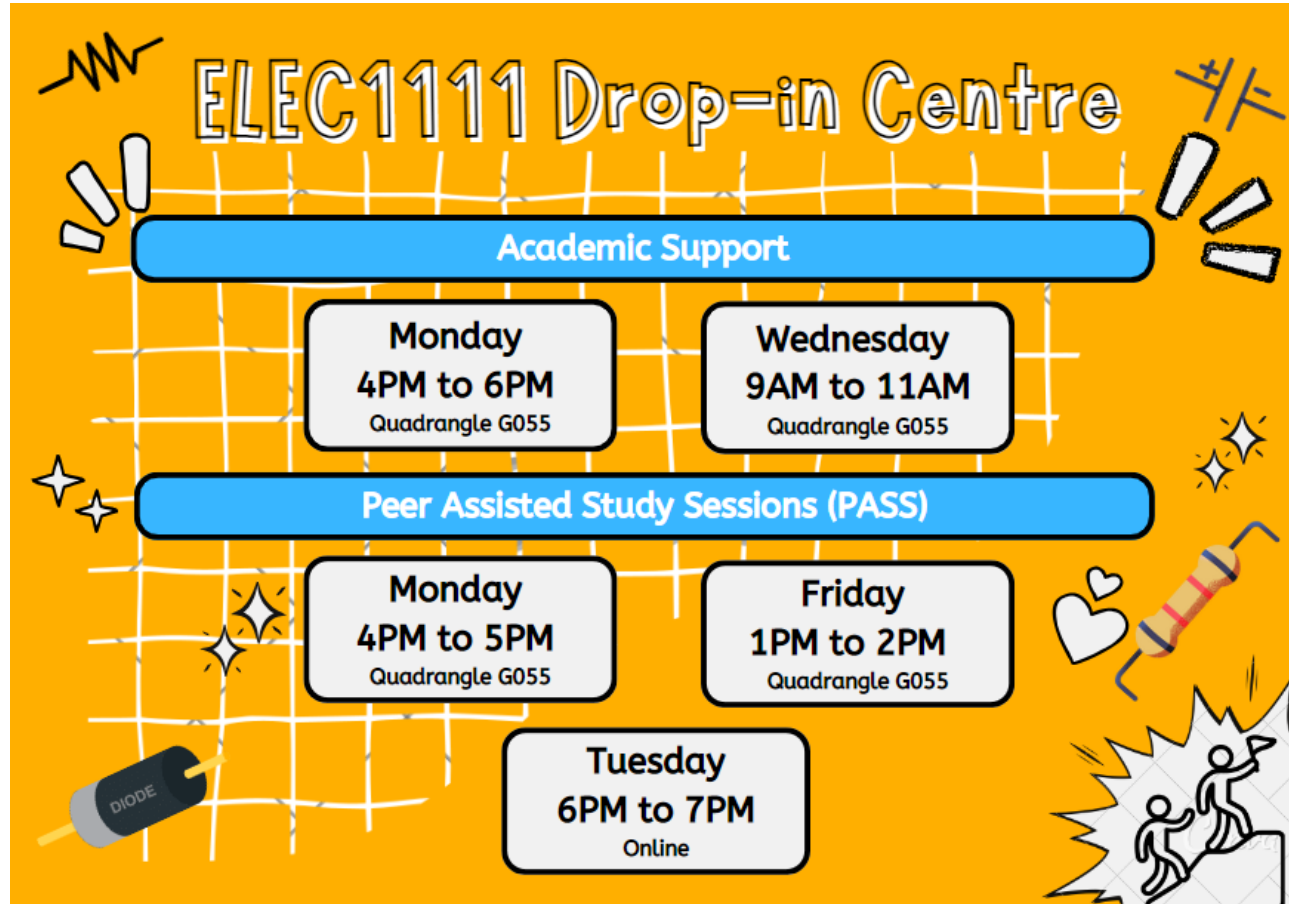
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UNSW
SYDNEY

Ask Away!

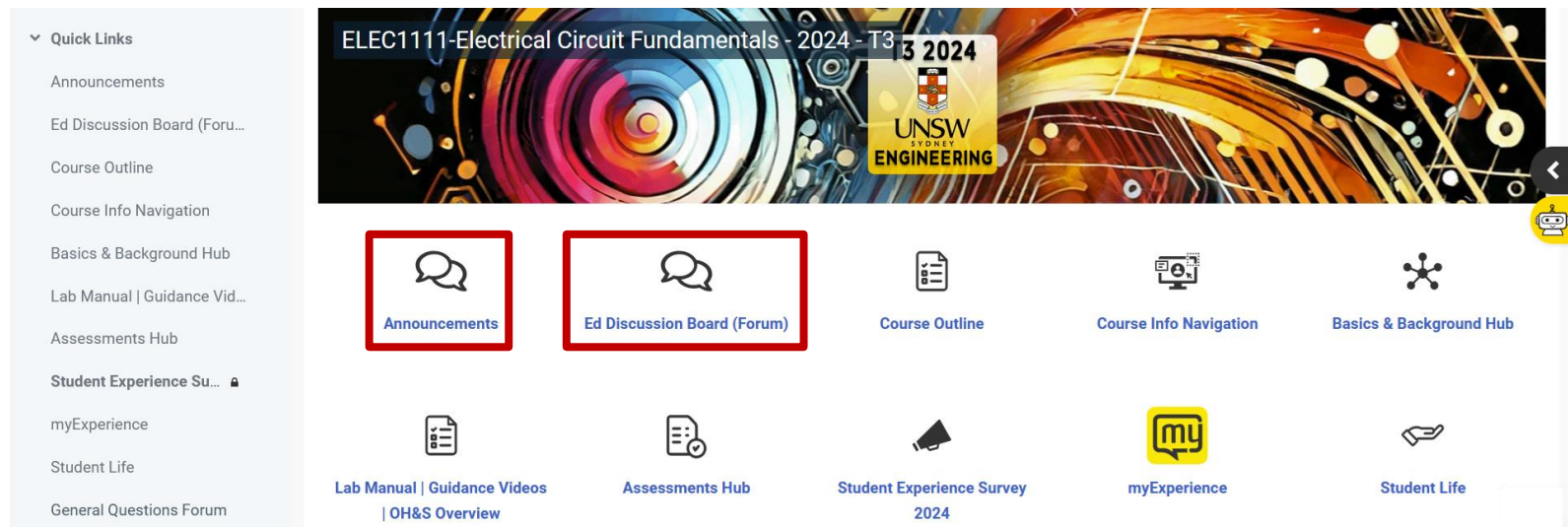
Drop-in centre



Online Learning System

Moodle

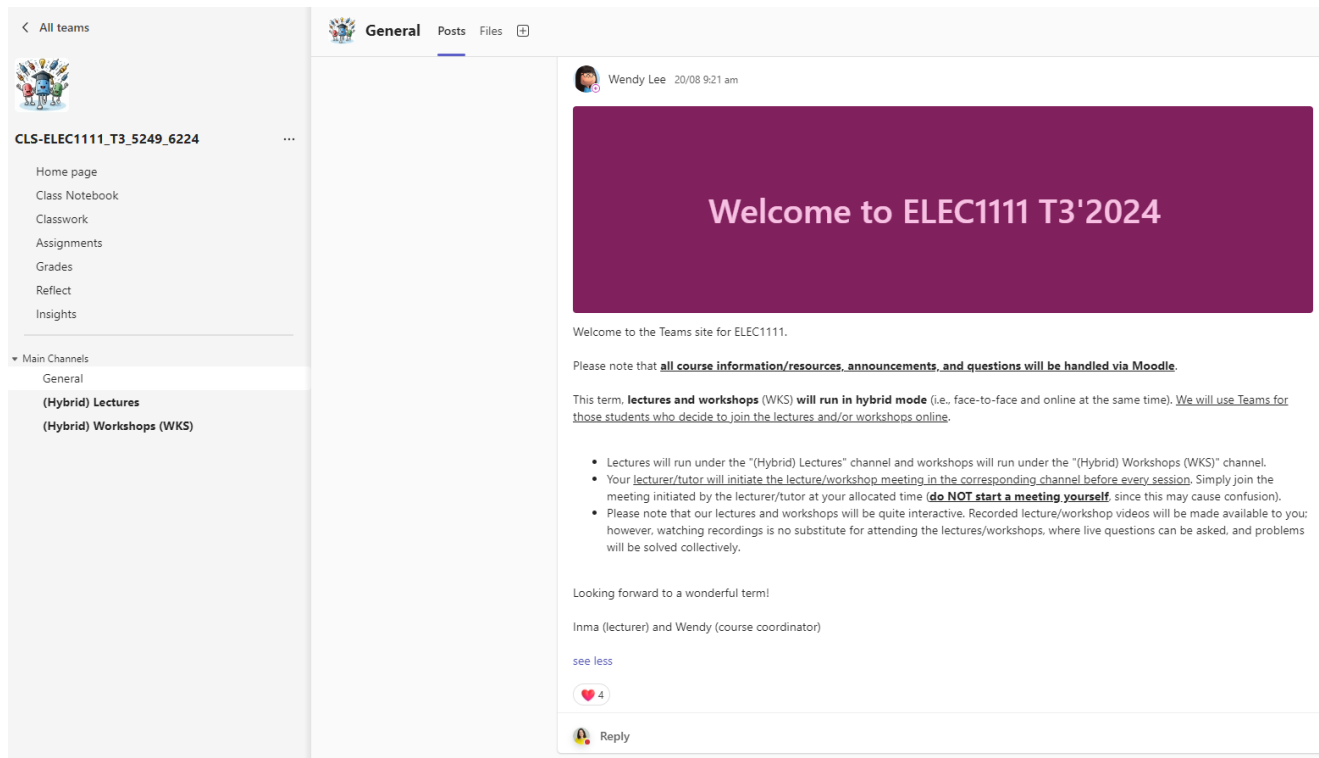
- Automatic access after enrolment in ELEC1111.
- All course materials will be available and all marks will be posted.
- **All official announcements will be posted on Course Announcements** (you will receive an email for each announcement).
- An Ed Discussion Board (<https://edstem.org/au/courses/19021/discussion/>) will be the main forum to ask questions pertaining course.



Online Learning System

Teams

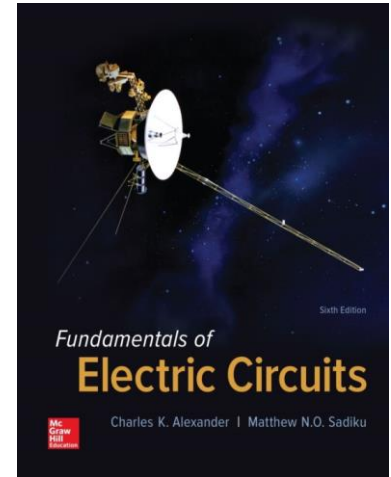
- Automatic access after enrolment in ELEC1111.
- Meetings for hybrid lectures/workshops (WKS) will be initiated by lecturer/tutor.
- Use **Ed Discussion Board** to ask questions, not Teams (not monitored).



Course Resources

Prescribed Textbook

- C. K. Alexander and M. N. O. Sadiku, *Fundamentals of Electric Circuits*, 7th ed., New York, NY, USA: McGraw-Hill, 2017.
- Where to get it:
 - UNSW Library.
 - UNSW Bookshop (print and electronic versions).
 - Upgrading your version of McGraw-Hill Connect to get your Smart Book (\$40):
Access via Moodle and follow instructions in “Instructions on how to obtain the e-book at a discounted rate (\$40) .pdf” under “General Course Resources Hub -> Textbook”.



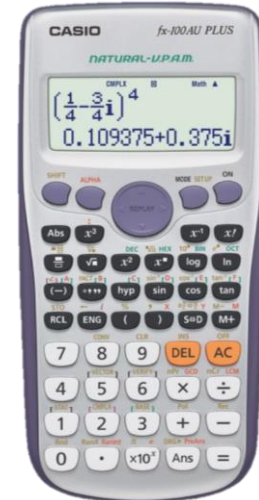
Course Resources

Calculator

- UNSW approved exam calculators:

<https://student.unsw.edu.au/exam-approved-calculators-and-computers>

- Recommendation: choose one with complex numbers computation capability.
 - “For the final exam (and your whole degree) get a Casio fx100au or Casio fx100au-plus. You just type in complex numbers straight in. It saves so much time. E.g. you can type $(1 + j)/(3 \text{ at } 60 \text{ degrees})$ and it will just figure out the answer in polar or rectangular form. If you don't get such a calculator, you will not do well. Learn how to use its functions prior to the exam.”
- Life saver and long-term investment.



Course Delivery

Lecture

- 2 x 2-hour sessions per week covering course topics and problem-solving.
- Hybrid (both in person and online), attendance very strongly recommended

Lab

- 1 hour of pre-lab per week. Pre-lab to be completed before attending the practical session independently or during pre-lab hour
- 2-hour practical session per week where you will **work with a partner. In-person, you must attend and attempt 6/7 labs** (Introductory Lab excluded).
- Lab exam: 1-hour **individual lab exam, in week 9. In-person, you must get a 50/100.**

Workshop

- 1.5-hour session per week where a tutor will solve problems with your inputs and help.
- In-person, attendance very strongly recommended

Supplementary Resources

- Solved examples (questions with detailed written solutions) for you to practice at your own pace.
- Independent tutorials (questions with video solutions) for you to practice at your own pace

NOTE: Everything starting in **Week 1**. Lecture notes without solutions will be posted in Moodle before the Wednesday lecture. Lecture notes with solutions will be posted after we finish the topic.

Course Planning

Week	Topic	Lectures		Workshops	Labs	Quizzes
Week 1	Topic 1	Wed	Introduction, Circuit Basics (current,voltage)	WKS 1	Introductory lab	Quiz 1
		Thurs	Circuit basics (power,energy) & basic elements (sources,Rs,diodes)			
Week 2	Topic 2	Wed	KCL, KVL, Series Parallel circuits, dividers	WKS 2	Lab 1 <i>Familiarisation</i>	Quiz 2
		Thurs	Nodal and Mesh Analysis			
Week 3	Topic 3	Wed	Circuit theorems - Superposition, Source Transf. and Thevenin	WKS 3	Lab 2 <i>Basic elements</i>	Quiz 3
		Thurs	Circuit theorems - Norton, Max. Power Transfer			
Week 4	Topic 4	Wed	Capacitors and RC Circuit Natural Response	WKS 4	Lab 3 <i>V-I characteristics & basic laws</i>	Quiz 4
		Thurs	RC Circuit Step Response			
Week 5	Topic 5	Wed	Inductors and RL Circuit Natural Response	WKS 5	Lab 4 <i>Circuit theorems</i>	Quiz 5
		Thurs	Mid-term exam , RL Circuit Step Response (recorded)			
Week 6					Open labs	
Week 7	Topic 6	Wed	Operational Amplifiers	WKS 6	Lab 5 <i>First order circuits</i>	Quiz 6
		Thurs	Operational Amplifiers			
Week 8	Topic 7	Wed	AC Analysis I - Phasors and Impedance	WKS 7	Lab 6 <i>Operational Amplifiers</i>	Quiz 7
		Thurs	AC Analysis I - Phasors and Impedance			
Week 9	Topic 8	Wed	AC Analysis II - Circuit Theorems & AC Op-Amps	WKS 8	Lab exam	Quiz 8
		Thurs	AC Analysis II - Circuit Theorems & AC Op-Amps			
Week 10	Topic 9	Wed	AC Power	WKS 9	Lab 7 <i>AC circuits</i>	Quiz 9
		Thurs	AC Power, Revision (recorded)			

Laboratories

- All lab resources (including manual) under “Laboratory Manual, Guidance Videos and OH&S”.
- Electrical Engineering OH&S module and submission (**compulsory**)
 - Self-enrolment module.
 - Enrolment key is **elec mood**.
 - Complete the module.
 - Submit the certificate before the first lab.
 - If you completed the module in the past, upload the old certificate.

Laboratory Manual, Guidance Videos and OH&S

In this section you will find essential resources for succeeding in the laboratory.

OH&S Module (compulsory)

 [Electrical Engineering OH&S Module](#)

Please complete the module on Occupational Health and Safety before your first lab. The enrolment key is **elec mood**

 [OH&S Module Certificate Submission](#)

Submit your OH&S certificates here before attending the first lab.



Laboratories

- **Lab component kit + prototyping board (breadboard)**
 - Purchase online via <https://recharge.it.unsw.edu.au> (see announcement for detailed steps).
 - Collect from EE School Workshop office (room G15, ground floor, G17 building), if possible before the Introductory Lab in week 1.
- **Goggles** are recommended and **mandatory to wear in Lab 6.**



Note: You can buy goggles from the workshop.



- **Remember:**
 - To wear enclosed shoes, no thongs!
 - Listen to lab announcements carefully.
 - Follow the instructions.
 - **Do not touch components when the circuit is energised...they can be quite hot.**



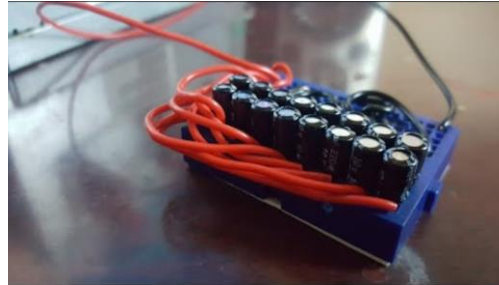
The temperature of the resistor is 41.4°C
This is something that may happen during Lab 4.

Laboratories

- It is not usual but...if you smell something “funny” coming from your circuit, **SWITCH ITS SUPPLY OFF!**

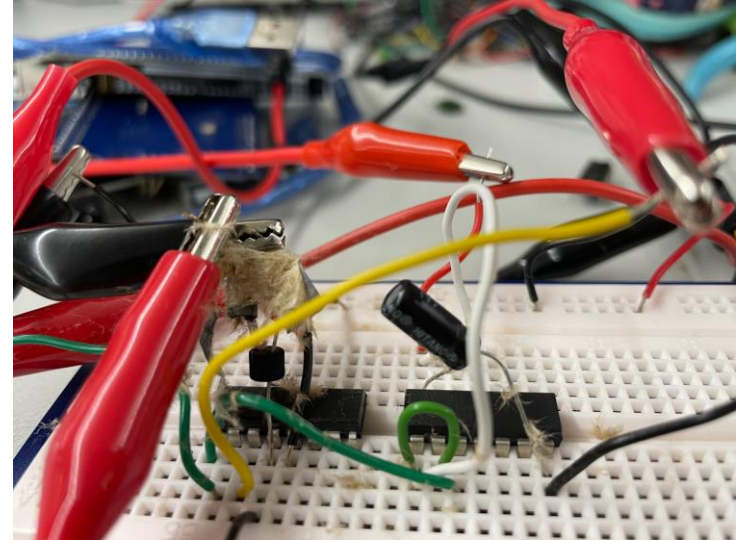


<https://youtu.be/rr7bPmGTQUk>

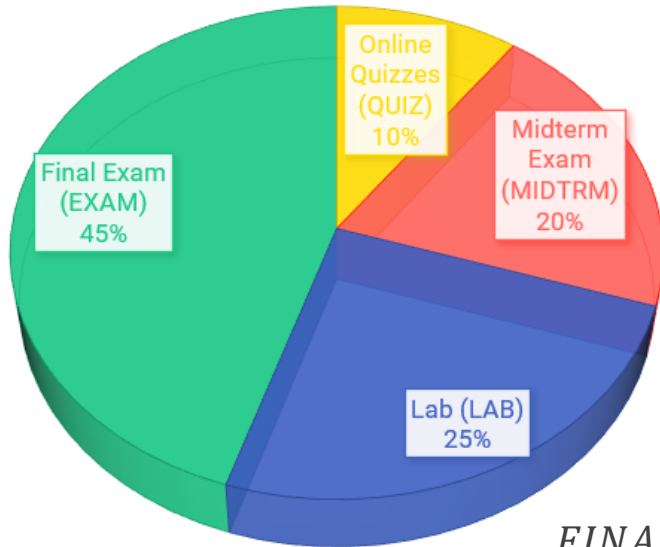


<https://youtu.be/NOVOON4IJog>

Electrolytic capacitors may explode if connected with the wrong polarity or due to overvoltages!



Assessment



$$FINAL = 0.45 \cdot EXAM + 0.25 \cdot LAB + 0.2 \cdot MIDTRM + 0.1 \cdot QUIZ$$

$$LAB = \left(\frac{0.15}{7} \cdot \sum_1^7 Lab_i + 0.1 \cdot Lab_{exam} \right) \cdot \frac{100}{25}$$

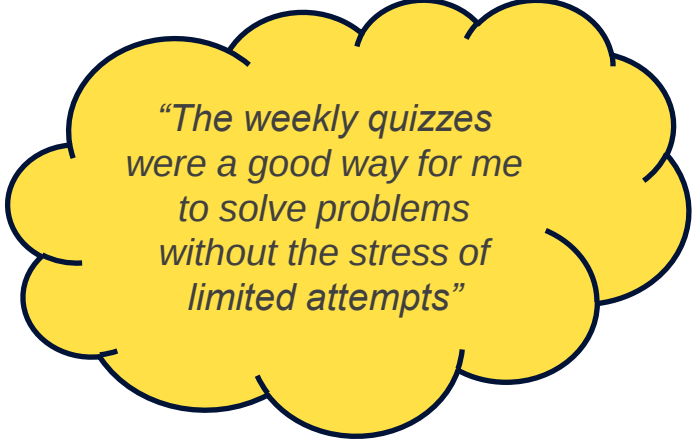
$$MIDTRM = \left(0.175 \cdot Exam + 0.025 \cdot Exam_{self-marking} \right) \cdot \frac{100}{20}$$

$$QUIZ = \frac{1}{9} \cdot \sum_1^9 Quiz_i$$

Note: Repeating students are NOT exempt from any of these assessment items.

Online Quizzes

- One online quiz for each topic, starting from Week 1.
 - Useful to evaluate your progress on each topic.
 - Module opens from Wed 11:59pm to the next Wed 11:59pm.
 - 5 questions, infinite attempts, immediate right/wrong feedback.
 - Detailed feedback provided after due date (on Wednesday 11:59pm).
- The online quizzes mark will be the **average** mark of the 9 quizzes.
- The online quizzes mark accounts for **10%** of the total course mark.



*“The weekly quizzes
were a good way for me
to solve problems
without the stress of
limited attempts”*

Mid-term Exam

- Face-to-face exam in Week 5.
 - Exam will take place during the Thursday lecture in week 5 (10th October @ 9am).
 - Exact rooms will be announced on Moodle closer to the date.
- Questions may be drawn from any material already covered in the course schedule up to week 4 (included).
- The mid-term exam mark accounts for **20%** of the total course mark.
 - 17.5% from the mark you receive from the markers.
 - 2.5% from self-marking task.
 - You will be asked to use the exam solution and rubric to self-evaluate your answer booklet.
 - Useful **to better understand your strengths and weaknesses**.



"I felt like that the midterm self marking activity really helped put into perspective what I needed to work on"

Laboratory assessment and exam

- LAB assessment consist of:
 - Weekly labs (7 + Introductory Lab), and
 - Lab exam.
- The LAB mark accounts for **25%** of the total course mark:
 - **15%** from weekly labs, and
 - **10%** from lab exam.
- Weekly labs starts in week 1 with the Introductory Lab:
 - Working in pairs is strongly encouraged.
 - The final mark of the weekly labs is the average mark of the 7 labs.
- Lab exam:
 - Individual, during week 9.

"The hands-on approach in the labs allowed me to see and understand the behaviours' that I had learnt about and calculated."

Hurdle

- Must attend at least 6 out of 7 labs (introductory lab excluded).

AND

- Pass lab exam (Week 9).



FAIL THE LAB!
FAIL THE COURSE !

Laboratory assessment

- Each weekly lab consists of 2 parts:
 - Analytical questions (pre-lab)
 - Roughly 1 hour of work (independent or on first hour of lab session)
 - Accounts for 20% of the weekly lab mark.
 - Video-solutions available.
 - Lab experiments (laboratory part)
 - 2 hours of work.
 - Accounts for 80% of the weekly lab mark.

$$Lab_i = 0.2 \cdot Analytical_i + 0.8 \cdot Practical_i$$

UNSW

A.1 Given the solar model and parameters shown in Figure 3.2, calculate the maximum, voltage available for each cell, V_{oc} . Note that the open circuit voltage, V_{oc} is the voltage across the solar cell when the current through it is zero (i.e., when there is nothing connected to terminals a-b) and can be analytically calculated as:

$$V_{oc} = n V_T \ln \left(\frac{I_L}{I_0} + 1 \right)$$

Where n is the diode ideality factor; V_T is the thermal voltage, which, at 300K (27°C) equals 25.86 mV; I_L is the light-generated current; and I_0 is the diode saturation current.

Answer: $V_{occell} = 0.61$ V

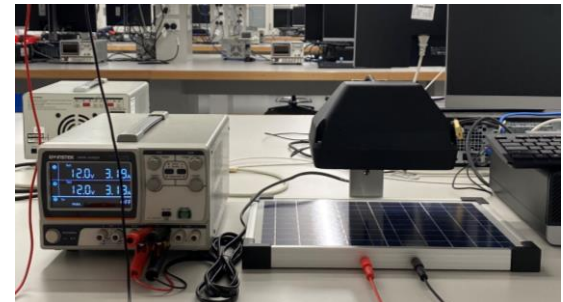
A.2 Knowing that the solar panel has 36 solar cells connected in series (as can be seen in Figure 3.1), calculate the open circuit voltage of the solar panel. Compare your calculated value with the value provided by the manufacturer in the specifications.

Answer: $V_{ocpanel} = 21.96$ V

A.3 Calculate the voltage across the fan and the 10 kΩ resistor when the output voltage of the solar panel is 12.5 V.

Answer: $V_{fan} = 12.5$ V; $V_{10k} = 10.8$ V.

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Laboratory assessment

Analytical part (pre-lab) Marking criteria (whole pre-lab)

Grade	Criteria
Excellent (1/1)	Work completed with correct answers
Average (0.5/1)	Work partially completed or with few incorrect answers
Zero (0/1)	Work not attempted or <50% complete
Absent (0/1)	No show

Lab experiments Marking criteria (per question)

Grade	Criteria
Excellent (1/1)	0 mistakes
Good (0.75/1)	1-2 minor mistakes
Average (0.5/1)	3-4 minor mistakes, or 1 major mistake
Poor (0.25/1)	4-5 minor mistakes, or 2 major mistakes
Zero (0/1)	+5 minor mistakes, or +2 major mistakes
Absent (0/1)	No show

Minor mistake: value with wrong polarity/direction, wrong scale, wrong/missing unit, unappropriated measurement precision, etc.

Major mistake: wrong setup, wrong measurement, etc.

$$Practical_i = \frac{1}{\#questions} \cdot \sum_1^{\#} Grades$$

Final Exam

- Face-to-face exam in examination period.
- Questions may be drawn from any aspect of the course.
- You must achieve a minimum of 40 marks out of 100 to pass the course.
- The final exam mark accounts for **45%** of the total course mark.



Assessment – What if...

- ...I cannot sit the midterm exam, or the lab exam, or the final exam?
 - [Special consideration](#) is required to sit an exam another day.
- ...I missed the midterm's self-marking deadline?
 - You get a 0 in the activity (unless you are granted [special consideration](#)).
- ...I missed a lab?
 - Fill out the [catch-up form](#), so you can be allocated to another session.
 - If you miss the catch-up session, your maximum mark for that lab will be 50 (once you attend and complete the lab).
 - Apply to [special consideration](#) if it is your 4th+ time requesting a catch-up. Otherwise, we will not process your request.
- ...a quiz?
 - If you were not able to complete the quiz during the whole week, [special consideration](#) is required.

Need Help?

Ask me during/
after lectures or
send me an email

Attend the
ELEC1111 Drop-in
Centre

Ask your tutor
during workshops

Post questions on
Ed Discussion
Forum

Ask your lab
demos during your
labs

Attend ELSOC's
Crash Courses

REMEMBER

We also have other
duties and need to have
a work-life balance as
well.

We will get back to you
as soon as possible,
mostly during working
hours.

DON'T WAIT !

Student Support | I Need Help With...

My Feelings and Mental Health Managing Low Mood, Unusual Feelings & Depression	 Mental Health Connect student.unsw.edu.au/counselling Telehealth	 In Australia Call Afterhours UNSW Mental Health Support Line 1300 787 026 5pm-9am
	 Mind HUB student.unsw.edu.au/mind-hub Online Self-Help Resources	 Outside Australia Afterhours 24-hour Medibank Hotline +61 (2) 8905 0307
Uni and Life Pressure Stress, Financial, Visas, Accommodation & More	 Student Support Indigenous Student Support	– student.unsw.edu.au/advisors – nura-gili-centre-indigenous-programs
Reporting Sexual Assault/Harassment	 Equity Diversity and Inclusion (EDI)	– edi.unsw.edu.au/sexual-misconduct
Educational Adjustments To Manage my Studies and Disability / Health Condition	 Equitable Learning Service (ELS)	– student.unsw.edu.au/els
Academic and Study Skills	 Academic Language Skills	– student.unsw.edu.au/skills
Special Consideration Because Life Impacts our Studies and Exams	 Special Consideration	– student.unsw.edu.au/special-consideration

Week 1: To-do List

- Read the course outline carefully.
- Check out Moodle and Teams.
- Complete the pre-course survey.
- Complete OH&S module and submit the certificate.
- Have a look at the Introductory Lab slides.
- Read the preface of the lab manual and watch the lab guidance videos.
- Read and complete the analytical part of Lab 1 before your allocated lab session in week 2 (to be found in lab manual).
- Buy and collect lab kit and breadboard in EEG15.

Questions?

